

Design Proposal: The Mach-Zehnder Interferometer

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1. Introduction

The Mach-Zehnder interferometer (MZI) is a versatile photonic device that can be employed in both research and the development of larger devices such as optical fiber sensors [1]. Consisting of a single input split into two branches and then recombined, the purpose of the MZI is to split light into two paths and introduce a phase shift in one of the paths. [2] This results in an interference pattern at the output that can be used for various optical applications. The objective of my design is to evaluate the effect of varying path length difference on the free spectral range (FSR) of the MZI.

2. Theory

It is important to first establish important relevant theory when it comes to analyzing the output of the MZIs that will be included in my design. The MZIs in my layout will all contain the same three sections: a splitting y-branch, two waveguides of the same material properties, width and height but differing path length, and a combining y-branch. An example of this can be seen in Figure 1.

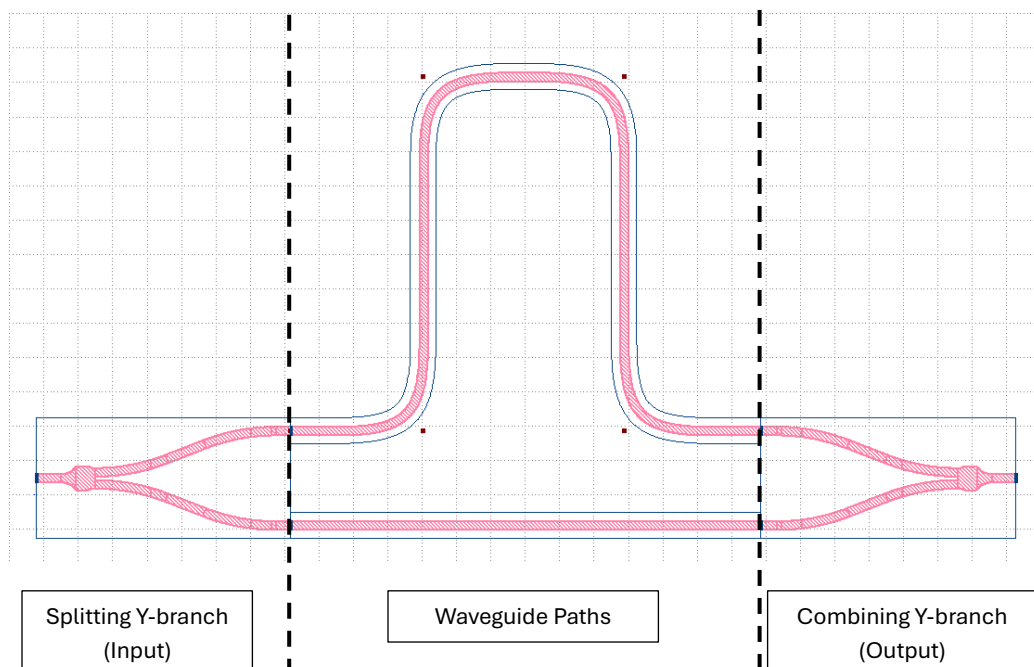


Fig. 1. Unbalanced MZI with differing waveguide path length, constructed in *KLayout* [3].

Two important values when it comes to such an interferometer are the light intensity I and the electric field E .

For a lossless unbalanced MZI, which is what will be looked at in this report, the output electric field E_o can be determined from (1), where E_{o1} and E_{o2} are the electric field outputs for each of the waveguides and β_1 and β_2 are the propagation constants for said waveguides [2].

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2} (e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}) \quad (1)$$

By taking the magnitude squared of the electric fields, it is possible to convert (1) into an expression representing the light intensities in different parts of the MZI [2].

$$I_o = \frac{I_i}{4} |e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}|^2 \quad (2)$$

This can finally be simplified to yield the expression [2]:

$$I_o = \frac{I_i}{2} [1 + \cos(\beta \Delta L)] \quad (3)$$

Equation 3 is known as the transfer function of the MZI, which displays a sinusoidal variation with wavelength. The period between the peak wavelengths of this function is known as the free spectral range FSR, and can be determined by the following expression [2]:

$$FSR = \frac{\lambda^2}{\Delta L n_g} \quad (4)$$

In (4), λ is the wavelength of the light travelling through the MZI while n_g is the group index.

Modelling and Simulation:

In my design, I employ waveguides with a width of 500nm and height 220nm. Additionally, they are strip-type waveguides and are quasi-TE polarized.

Firstly, I simulated the TE and TM mode profiles for a 1.55 μm optical stream in the described waveguide using *Lumerical MODE* [4]. These are represented by Figure 2 and Figure 3 respectively.

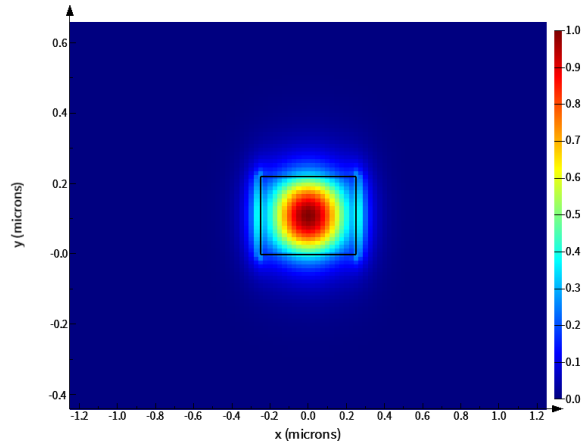


Fig. 2 Mode profile of the first TE mode.

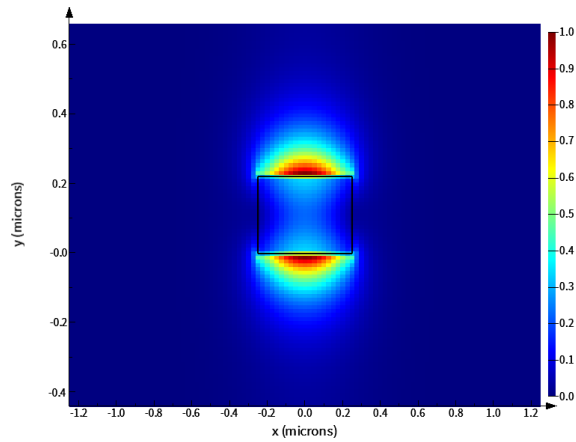


Fig. 3. Mode profile of the first TM mode.

Next, also using *Lumerical MODE* [4], I plotted the effective index of the waveguide against the wavelength of the light, which can be seen in Figure 4.

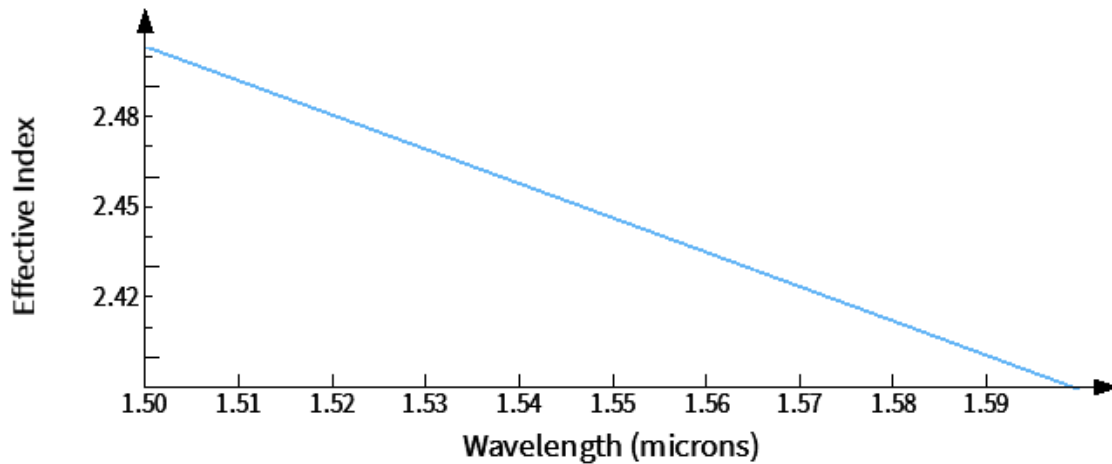


Fig. 4 Effective index of the waveguide versus length for the first TE mode.

I also plotted the group index of the waveguide against the wavelength as well, shown in Figure 5.

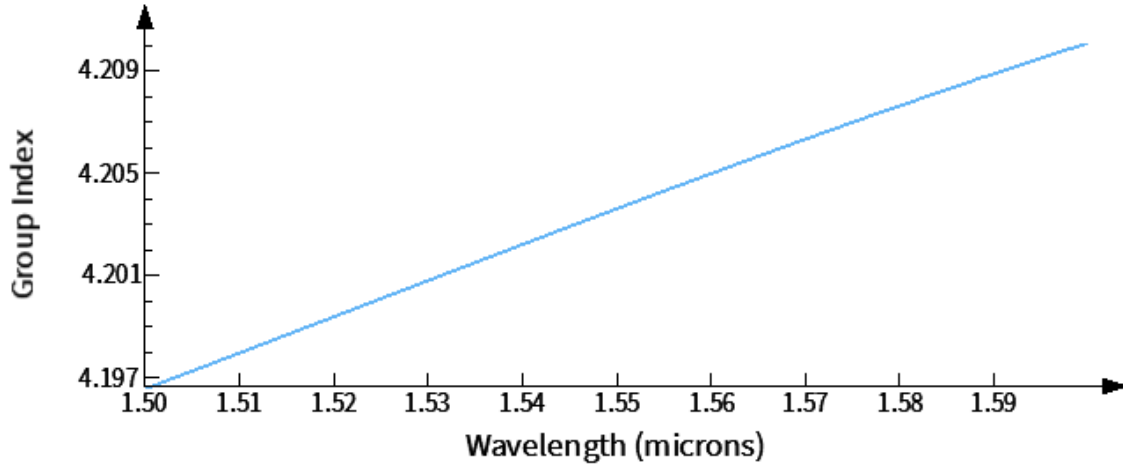


Fig. 5. Group index of the waveguide versus length for the first TE mode.

Once again using *Lumerical MODE* [4], I calculated the compact model of the waveguide which turned out to be:

$$n_{eff}(\lambda) = 2.44682 - 1.13352(\lambda - 1.55) - 0.041065(\lambda - 1.55) \quad (5)$$

For each MZI, we know that the wavelength λ is fixed at $1.55\mu\text{m}$ and the group index n_g , which was calculated using *Lumerical Interconnect* [5], is fixed at 4.20586. Therefore, it is possible to determine the expected FSR for each unbalanced MZI depending on their respective path length difference ΔL by using (4). This is shown in Table I.

TABLE I
PATH LENGTH DIFFERENCE VS FREE SPECTRAL RANGE

ΔL (μm)	FSR (nm)
100	5.712268
150	3.808179
200	2.856134
250	2.284907
300	1.904089
350	1.632077

This can be compared to the transmission spectrums of each MZI simulated in *Lumerical Interconnect* [5], which is shown in Figure 6.

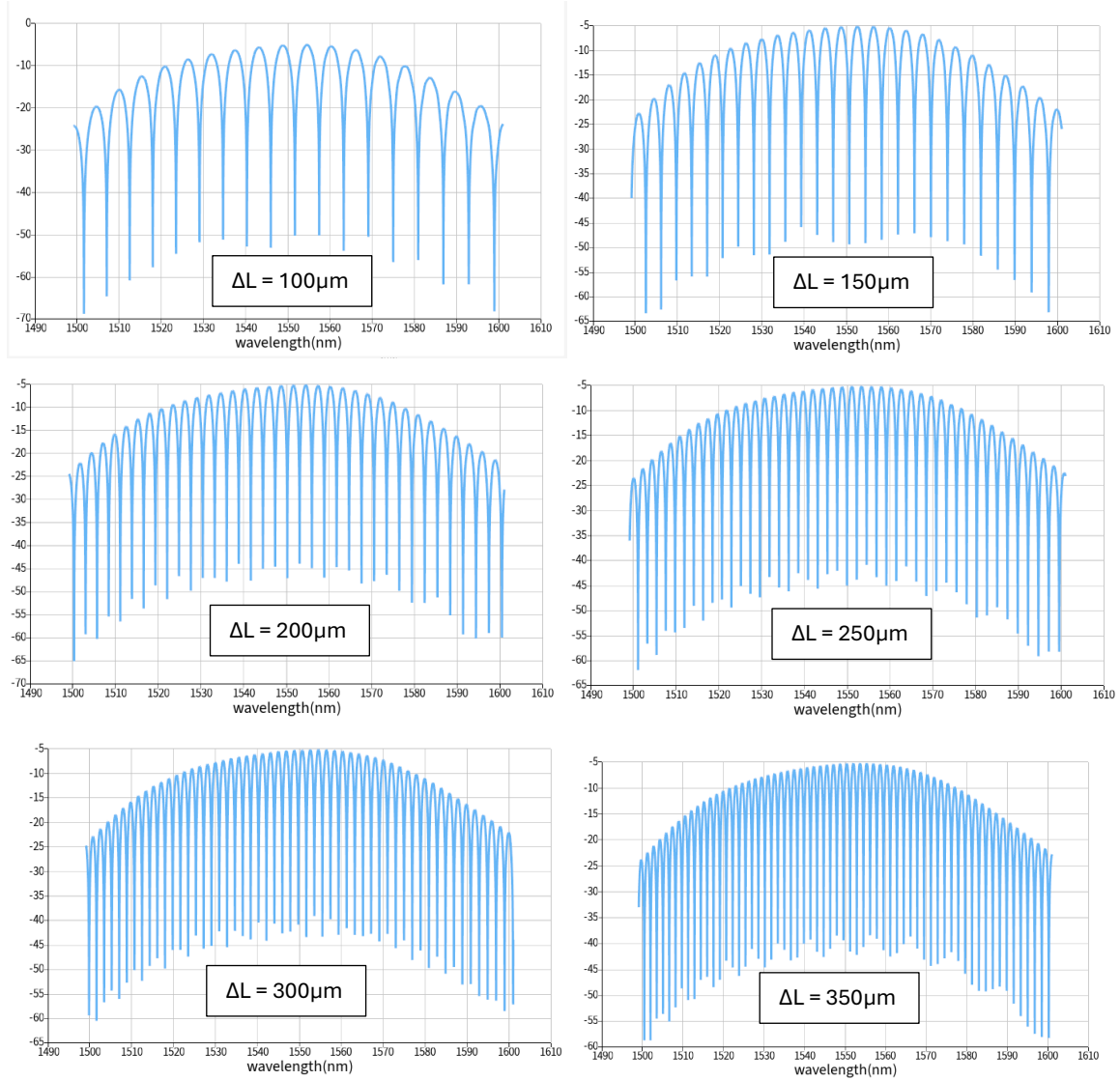


Fig. 6. Transmission spectrums for unbalanced MZIs with varying path length difference.

Further Plans:

Once the layout is fabricated and I receive the experimental data, I will rearrange (4) to determine the experimental group index n_g :

$$n_g = \frac{\lambda^2}{\Delta L(FSR)} \quad (6)$$

We can test this equation using the FSR range seen in Figure 6 for $\Delta L = 100\mu\text{m}$, which is approximately 5.7nm.

$$n_g = \frac{(1.55\mu\text{m})^2}{(100\mu\text{m})(0.0057\mu\text{m})} = 4.21491 \quad (7)$$

This is similar to the calculated n_g value of 4.20586, proving that this equation will be useful for comparing the experimental data I will receive to the simulated data I already calculated.

References

- [1] “Optica Publishing Group,” *Optica.org*, 2025. <https://opg.optica.org/oe/fulltext.cfm?uri=oe-20-10-11109> (accessed Oct. 02, 2026).
- [2] Lukas Chrostowski and M. Hochberg, *Silicon photonics design : from devices to systems*. Cambridge: Cambridge University Press, 2016.
- [3] *KLayout* software. <https://www.klayout.de/>
- [4] *Lumerical MODE* software <https://ansys.synopsys.com/products/optics/mode>
- [5] *Lumerical Interconnect* software <https://ansys.synopsys.com/products/optics/interconnect>